

Catch my Eyebrow, Catch my Mind: Examining the Effect of Upper Facial Expressions on Emotional Recognition for VR Avatars

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ABSTRACT

Capturing the movement of the upper face in social VR was challenging due to the head-mounted display's occlusion, which hinders accurate emotional expression. In this paper, we conducted two user studies to examine the effect of upper facial expressions on users' emotional recognition ability with VR avatars. Study 1 found that compared with eyelids, restricting the movement of eyebrows would yield greater difficulty for the users to recognize the expressed emotions of 3D avatars. Study 2 further tested different movement amplitudes of the upper facial expressions and results showed that the most realistic movement amplitude was optimal when balancing the tradeoff between emotion recognition accuracy and the Uncanny Valley effect.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality

1 INTRODUCTION

In social virtual reality (VR), avatars bridge users' social interactions and emotional expressions. Although recent VR headsets have integrated facial-tracking sensors, reconstructing accurate facial expressions is still a challenge because *Upper Facial Expression* (UFE, expressed by the eye, forehead, and associated muscles) is usually masked by the head-mounted display (HMD). Currently, eye-tracking devices inside the HMD capture eye movements to generate the virtual avatars' UFE. A more detailed understanding of how avatar UFE affects emotional recognition is essential for natural and expressive VR communication.

Prior work [1, 3, 5] provided insights into the contributions of facial features to process and recognize different facial expressions. However, most studies used static and partial facial images as materials, which were unlikely to occur in social VR situations. To fill the gap, we conducted two studies to directly examine the emotional recognition of avatars in a VR-mediated experiment.

In our studies, we chose avatars that were similar in style to those used in popular social VR applications (e.g., VRChat). We focused on the factors related to the motion of the UFE and examined the effect on emotional recognition of avatars. In Study 1, we refined the UFE to eyelid motion (e.g., eye widened or narrowed) and eyebrow motion (e.g., eyebrows raised or lowered) to explore the optimal UFE motion pattern. Results showed that eyebrow motion contributed to

higher recognition accuracy and less Uncanny Valley effect, while no such effect was found for eyelids, indicating the important role of eyebrows motion in expressing avatars' emotions.

Our results revealed that the accuracy of emotion recognition decreased without UFE, we completed the UFE in Study 2 and investigated the optimal motion amplitude of UFE for emotion recognition. Results showed that the most realistic (100) motion amplitude was optimal when balancing the trade-off between ease of emotion recognition and the Uncanny Valley effect.

2 STUDY 1: EFFECT OF UFE IN EMOTION RECOGNITION TASK

We recruited 22 participants who are college students (12 males, *Mean* age=24.2, *SD*=4.5), which ensured a statistic power > 0.8 ¹. We restricted the motion of eyelids and eyebrows separately when expressing the six basic emotions, i.e., happy, sad, angry, surprise, disgust and fear, see Figure 1a-1f), resulting in 2 (eyelids: Static, Dynamic) * 2 (eyebrows: Static, Dynamic) * 6 (emotions) = 24 conditions. All the UFE was composited with dynamic lower faces.

To generate the representative facial expressions, we decoded 6 emotions as facial unit combinations based on the previous facial coding system [4] using Blender and the Faceit add-on. We used shape keys in Unity to compile all facial units and took the value 100 as the endpoint of the facial expression. Then, we created animations by recording the process of shape keys going from 0 to 100. We designed and implemented an experimental platform in Unity version 2020.3.4 and used Oculus Quest 2 VR HMD.

The virtual environment consisted of a table and a chair with an avatar sitting on it. On each trial, an avatar appeared with facial animation, and then the animation played three times. After the animation, a 5-item questionnaire showed up. The first question asked participants to select the specific emotion that they perceived in the avatar's face from seven optional choices (labeled 6 basic emotions and the neutral one). The other four questions were designed to measure the Uncanny Valley effect through four semantic differential scales that recorded a 7-point value ranging from -3 to +3 [2]: Artificial-Natural (A-N), Reassuring-Eerie (R-E), Uninspiring-Spine-tingling (U-S), and Unattractive-Attractive (U-A).

2.1 Results

We analyzed the accuracy of emotional recognition and the Uncanny Valley effect scores using RM-ANOVA. A main effect of eyebrows on accuracy was found ($F_{1,21} = 14.52, p < .001$). The accuracy of dynamic eyebrows ($M = 0.61, SE = 0.02$) was higher than that of static eyebrows ($M = 0.51, SE = 0.02$). Surprisingly, whether or not the eyelid movement was disabled did not significantly affect the accuracy of emotional recognition ($F_{1,21} = 0.001, p = .98$).

For the Uncanny Valley effect, the effect of eyebrow movement was significant on the score of A-N, U-S, and U-A ($F_s > 15.96, ps <$

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¹<https://jakewestfall.shinyapps.io/pangea/>



Figure 1: Six basic emotions of male avatars used in Study 1 and 2.

.002). The mean difference score from dynamic minus static eyebrow was 0.57 ($SE = 0.14$), 0.52 ($SE = 0.12$), and 0.53 ($SE = 0.11$) for each scale. Findings implied that the dynamic eyebrows contributed to users' perceptions of more natural, spine-tingling, and attractive avatar expressions in VR. We also found that the effect of eyelid movement on the Uncanny Valley effect was significant ($F_s > 7.48, p_s < .02$), but in an unexpected way. That was, avatars with dynamic eyelids were perceived as less natural and eerier with the mean difference scores of -0.19 ($SE = 0.07$) and 0.18 (0.07) for A-N and R-E, respectively.

Results indicated that dynamic eyebrows played an important role in both improving emotion recognition and reducing the Uncanny Valley effect. However, the motion of eyelids should be handled with care in avatars, as we found eyelids had no effect on emotion recognition and had negative effects on naturalness and reassurance. Therefore, it is essential for cartoon avatars to have eyebrow movement but selective eyelid movement.

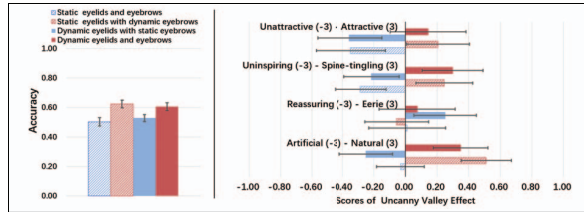


Figure 2: The accuracy rate of emotional recognition (left) and Uncanny Valley effect scores (right) in Study 1. Error bars depict $\pm 1SE$.

3 STUDY 2: EFFECT OF MOTION AMPLITUDE OF UFE IN EMOTION RECOGNITION TASK

We recruited 23 participants (7 males, $Mean$ age = 24.2, $SD = 3.61$) for Study 2 and excluded one subject because 23 out of 48 trials were not recorded.

We adopted the environment of Study 1, except for adapting the specific avatar materials and the questionnaire. We created avatars with four levels of UFE amplitude and six emotions, as the materials, resulting in 4 (amplitude: 75, 100, 125, 150) $\times$ 6 (emotions) = 24 conditions. The amplitude referred to the value of the endpoint of the facial expression. A large value meant a larger deformation during the animations from neutral to emotional expression. The first question of the questionnaire was modified to ask participants to rate the intensity of the emotion they perceived in the avatar's face. Participants were given the corresponding emotion label and asked to rate it from -3 (not at all) to +3 (tremendously). The other four questions measuring the Uncanny Valley effect were the same as in Study 1, except that they were changed to binary forced choices,

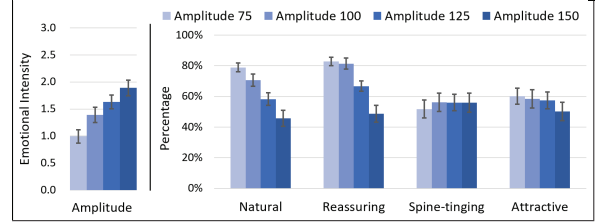


Figure 3: The emotional intensity score (left) and the percentage of positive words chosen for the Uncanny Valley effect (right) in Study 2. Error bars depict $\pm 1SE$.

making it easier for participants to choose the option that matched their feelings.

3.1 Results

The average emotional intensity was 0.99 ($SE = 0.13$), 1.39 ($SE = 0.14$), 1.63 ($SE = 0.13$), and 1.89 ($SE = 0.14$) for the increasing amplitude of the UFE, respectively. RM-ANOVA showed that emotional intensity increased linearly with amplitude ($F_{2,12,44.53} = 25.40, p < .001$). However, the Uncanny Valley effect also increased with motion amplitude, such that the percentages of choosing artificial and reassuring were higher for large amplitudes (e.g., 150, 125) than for small amplitudes ($p_s < .035$). Interestingly, no difference in the Uncanny Valley effect was observed between 75 and 100, while this increase in amplitude improved the emotional intensity of avatars. Thus, the 100 amplitude movement of the UFE was optimal when balancing the trade-off between ease of emotion recognition and the Uncanny Valley effect.

4 CONCLUSION

In this poster, we provided insights into the roles of UFE that can inform the future development of social VR technologies and devices. Through Study 1, we learned that dynamic eyebrows play an important role in both reducing the Uncanny Valley and increasing emotion recognition. Study 2 revealed the optimal parameter (100) that balances emotional expression and the Uncanny Valley effect. Our work was the first to investigate the amplitude and different parts of the UFE. It provided insights for future VR avatar design and face generation algorithm development.

ACKNOWLEDGMENTS

This work was supported by the Natural Science Foundation of China (NSFC) under Grant No. 62132010, and the grant from the Institute for Guo Qiang, Tsinghua University No. 2019GOG0003.

REFERENCES

- [1] M. Guarnera, P. Magnano, M. Pellerone, M. I. Cascio, V. Squatrito, and S. L. Bucchini. Facial expressions and the ability to recognize emotions from the eyes or mouth: A comparison among old adults, young adults, and children. *The Journal of genetic psychology*, 179(5):297–310, 2018.
- [2] C.-C. Ho and K. F. Macdorman. Revisiting the uncanny valley theory: Developing and validating an alternative to the godspeed indices. *Comput. Hum. Behav.*, 26:1508–1518, 2010.
- [3] S. Maher, T. Ekstrom, and Y. Chen. Greater perceptual sensitivity to happy facial expression. *Perception*, 43(12):1353–1364, 2014.
- [4] Y.-I. Tian, T. Kanade, and J. Cohn. Recognizing action units for facial expression analysis. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 23(2):97–115, Feb. 2001. Conference Name: IEEE Transactions on Pattern Analysis and Machine Intelligence. doi: 10.1109/34.908962
- [5] S. Zhang, X. Liu, X. Yang, Y. Shu, N. Liu, D. Zhang, and Y.-J. Liu. The influence of key facial features on recognition of emotion in cartoon faces. *Frontiers in psychology*, p. 3328, 2021.